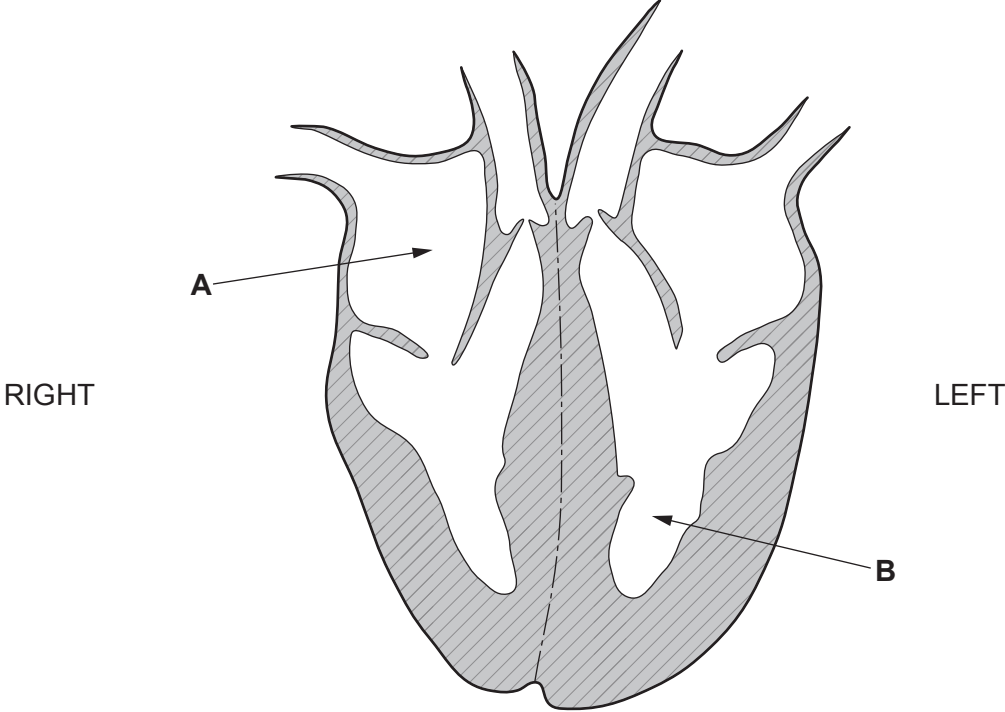


4. **Image 4.1** shows a section through a human heart.

Image 4.1



(a) (i) State the name of chambers **A** and **B** shown in **Image 4.1**. [1]

A

B

(ii) Humans rely on a double circulatory system. State what is meant by a double circulatory system. [2]

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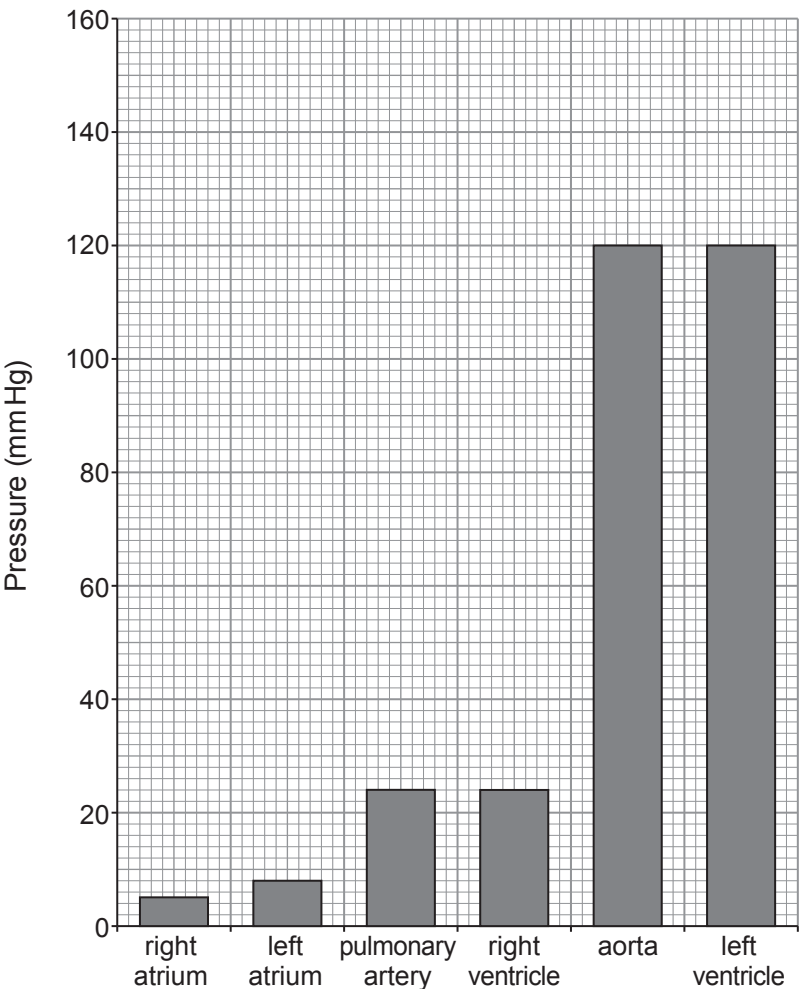
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(b) **Graph 4.2** shows the maximum pressure that occurs in the heart and associated blood vessels during the cardiac cycle.

Graph 4.2



- (i) I. Calculate the difference in the maximum pressure between the right ventricle and the left ventricle. [1]

Difference in pressure = mmHg

- II. State how the structure of the left ventricle causes this difference in pressure. [1]

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- (ii) Suggest why the maximum pressure in the aorta is the same as that in the left ventricle. [1]

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(c) Cardiovascular disease (CVD) is a general term for conditions affecting the heart or blood vessels. High blood pressure is one of the most important risk factors for CVD. Scientists carried out an investigation to test the following hypothesis:

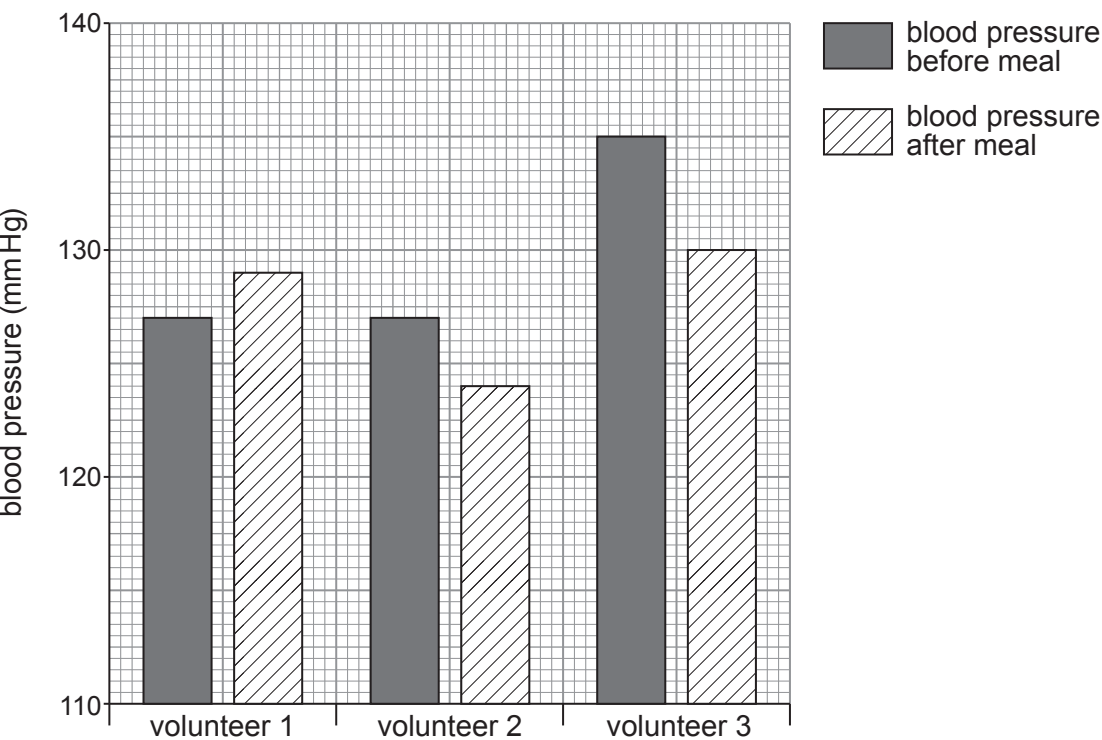
‘Eating food or drink that is high in nitrate causes blood vessels to widen, leading to a decrease in blood pressure.’

The investigation was carried out as follows:

- Three volunteers were each given a different meal.
- Volunteer 1: A salad made from vegetables that naturally contain almost no nitrate.
- Volunteer 2: A salad containing spinach and rocket, which have naturally high levels of nitrate.
- Volunteer 3: A portion of beetroot juice, calculated to contain exactly the same quantity of nitrate as the meal eaten by volunteer 2.
- The blood pressure of each volunteer was tested before and after their meal.

The results are shown in **Graph 4.3**.

Graph 4.3



State **two** conclusions which can be made from these results. [2]

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(d) Suggest why the widening of coronary arteries can lower the risk of a heart attack. [2]

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Examiner
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